

Internship Task – 2

Embedded Systems Lab Setup & Sensor Interfacing

MOTION DETECTION SECURITY SYSTEM USING ARDUINO UNO AND PIR SENSOR.

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ABSTRACT

This project presents a Motion Detection Security System using Arduino UNO and a PIR (Passive Infrared) Sensor. The system detects human motion and automatically turns ON an LED when movement is detected. The project demonstrates sensor interfacing, digital input processing, and real-time monitoring using an embedded system.

OBJECTIVE

The objective of this project is to interface a PIR motion sensor with Arduino UNO and detect movement in the surrounding environment. When motion is detected, an LED is turned ON and a notification message is displayed on the Serial Monitor.

COMPONENTS USED

- Arduino UNO
- PIR Motion Sensor
- LED
- Jumper Wires
- Wokwi Simulator

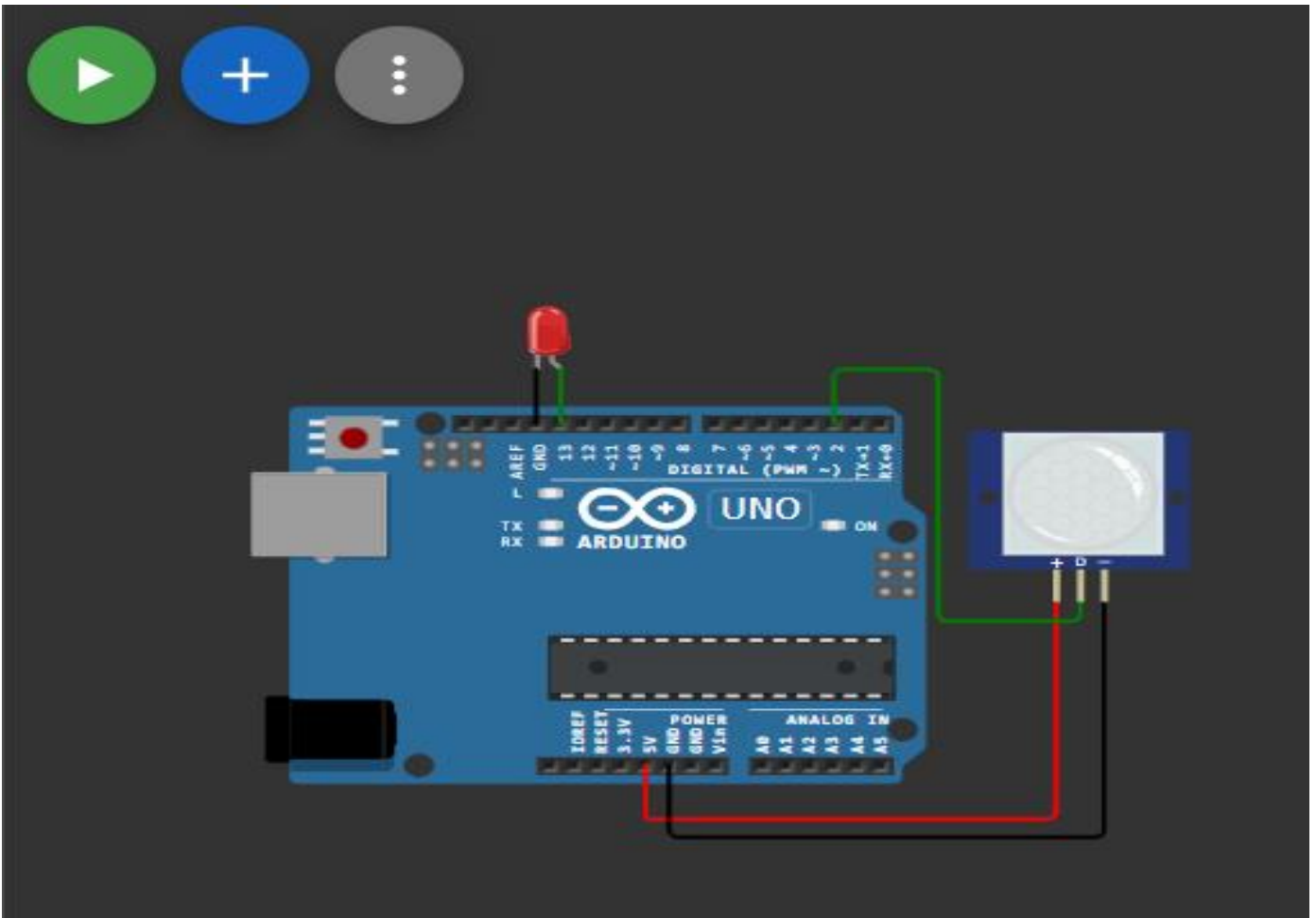
INTRODUCTION TO PIR SENSOR

PIR (Passive Infrared) Sensor is an electronic sensor that detects infrared radiation emitted by humans and animals. It is widely used in security systems, automatic lighting systems, and motion detection applications.

• Features of PIR Sensor

- Detects motion without physical contact
- Low power consumption
- Reliable and cost-effective
- Easy interfacing with microcontrollers

CIRCUIT DIAGRAM



CIRCUIT CONNECTIONS

• PIR Sensor Connections

- PIR VCC → Arduino 5V

- PIR GND → Arduino GND
- PIR OUT → Arduino Digital Pin 2

• LED Connections

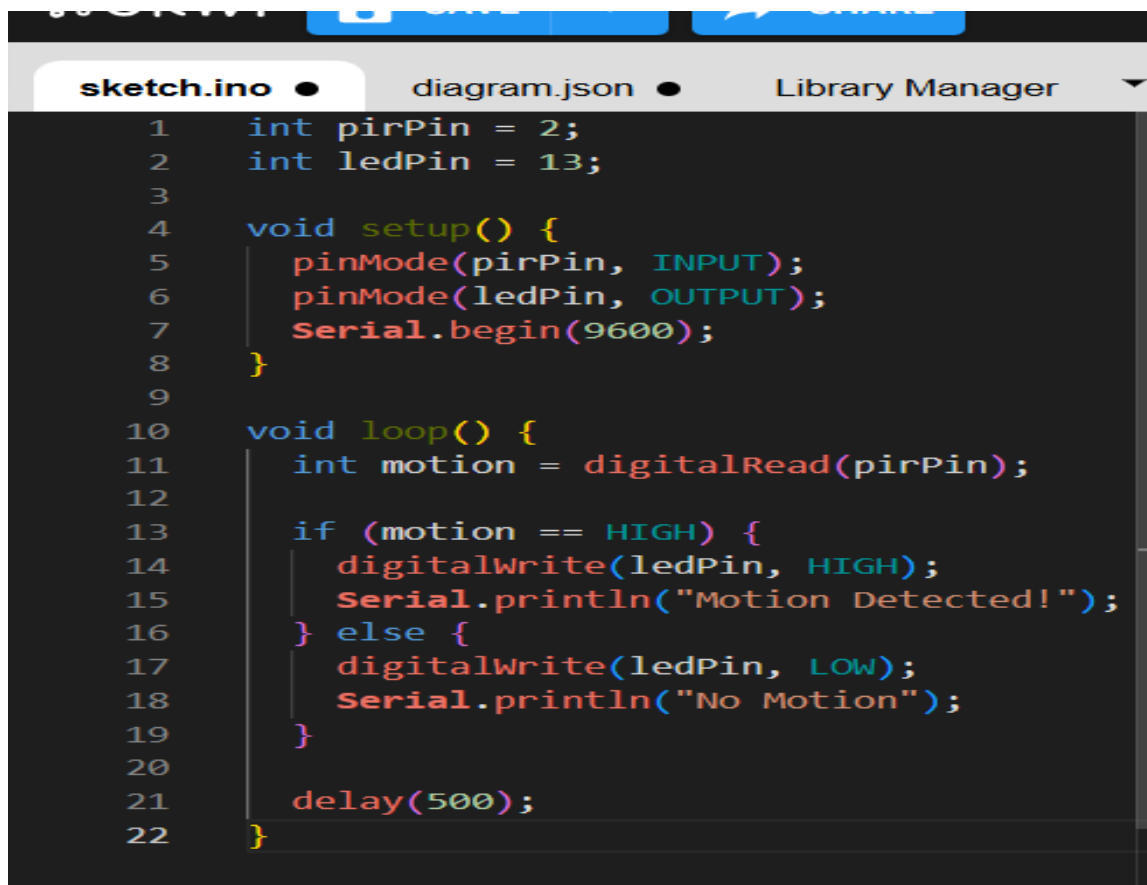
- LED Anode (+) → Arduino Digital Pin 13
- LED Cathode (-) → Arduino GND

WORKING PRINCIPLE

The PIR sensor continuously monitors infrared radiation in its surroundings. When a person moves in front of the sensor, the infrared radiation pattern changes and the PIR sensor generates a HIGH output signal.

The Arduino reads this signal through Digital Pin 2. If motion is detected, Arduino turns ON the LED and displays the message "Motion Detected!" on the Serial Monitor. When no motion is detected, the LED remains OFF and the Serial Monitor displays "No Motion".

ARDUINO PROGRAM



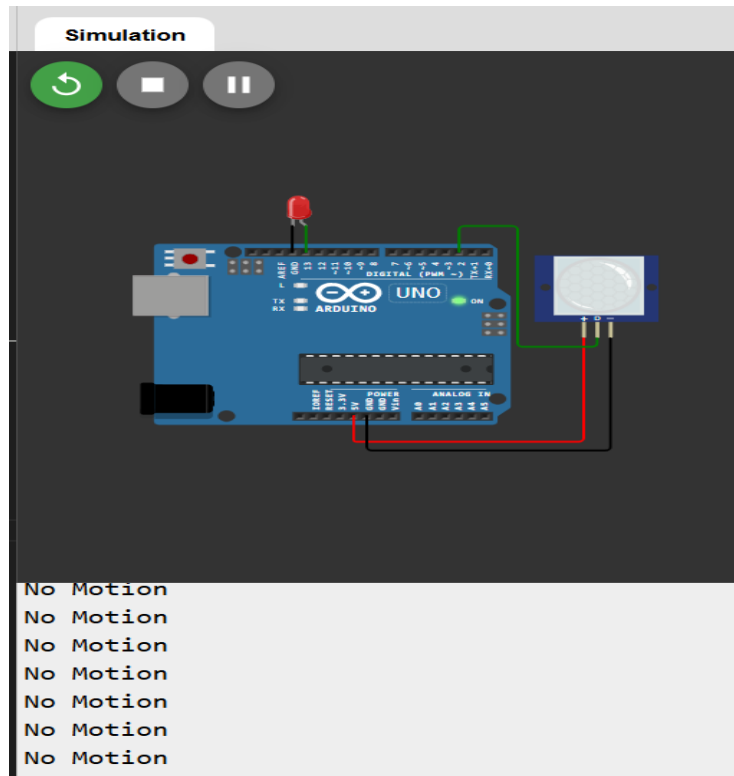
```
sketch.ino • diagram.json • Library Manager
1  int pirPin = 2;
2  int ledPin = 13;
3
4  void setup() {
5      pinMode(pirPin, INPUT);
6      pinMode(ledPin, OUTPUT);
7      Serial.begin(9600);
8  }
9
10 void loop() {
11     int motion = digitalRead(pirPin);
12
13     if (motion == HIGH) {
14         digitalWrite(ledPin, HIGH);
15         Serial.println("Motion Detected!");
16     } else {
17         digitalWrite(ledPin, LOW);
18         Serial.println("No Motion");
19     }
20
21     delay(500);
22 }
```

RESULTS AND OBSERVATIONS

Case 1: No Motion Detected :

Output: No Motion

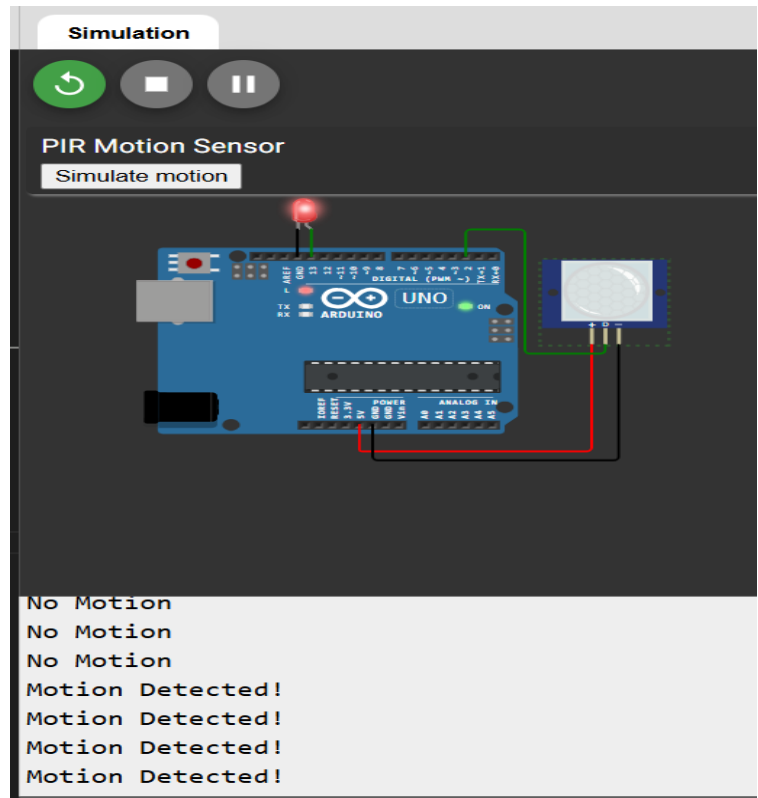
LED Status: OFF



Case 2: Motion Detected :

Output: Motion Detected!

LED Status: ON



APPLICATIONS

- Home Security Systems
- Motion Detection Alarms
- Automatic Lighting Systems
- Smart Homes
- Office Security Monitoring
- Energy Saving Systems

ADVANTAGES

- Easy Implementation
- Real-Time Motion Detection
- Low Power Consumption
- Reliable Operation

CONCLUSION

The Motion Detection Security System using Arduino UNO and PIR Sensor was successfully designed and simulated using the Wokwi platform. The system accurately detected motion and controlled the LED accordingly. This project provided practical experience in sensor interfacing and embedded system programming.

REFERENCES

- Arduino Official Documentation
- Wokwi Simulator Documentation
- PIR Sensor Datasheet
- Embedded Systems Fundamentals